

- 4 The early development of an animal involves divisions of the zygote and daughter cells by mitosis to form an embryo consisting of genetically identical cells.

Fig. 4.1 shows several cells at various stages of the cell cycle in an embryo of whitefish, *Coregonus artedii*.

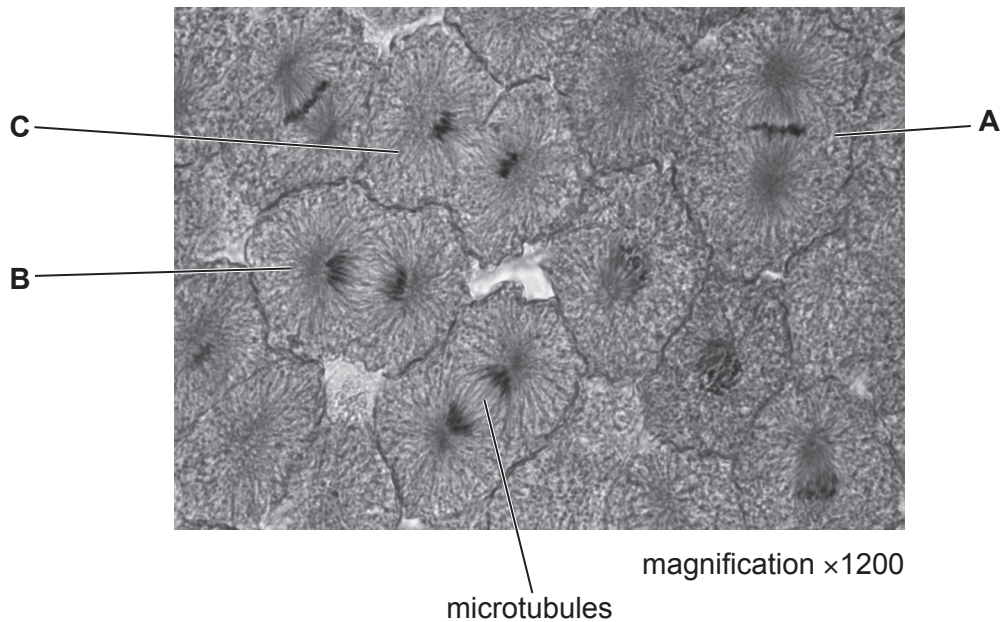


Fig. 4.1

- (a) (i) Name the stage of mitosis in cell **A** and in cell **B**, shown in Fig. 4.1.

A

B

[2]

- (ii) Fig. 4.1 shows microtubules in the cells that are dividing.

Describe the role of microtubules in mitosis.

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(iii) State what happens in cell **C**, shown in Fig. 4.1, until two new cells are formed.

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(b) The cells produced in the developing whitefish are genetically identical.

Identify **and** explain **two** events that occur during the cell cycle that lead to daughter cells being genetically identical.

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..... [3]

[Total: 11]